Naj bo $x, y \in A^*$. Pokaži da je enakost $xy = yx$

$xy = yx \iff \exists z \in A^* : x = z^n, y = z^m$

($\Leftarrow$) $z^n z^m = z^{n+m} = z^{m+n} = z^m z^n$

($\Rightarrow$)

$$x_1 \dots x_n y_1 \dots y_m = y_1 \dots y_m x_1 \dots x_n$$

BŠZS $n \geq m$

$$x_1 = y_1, \dots x_m = y_m = x_n$$

$$y_{m-1} = x_{m-1} = x_{n-1}$$

$$\vdots$$

$$y_1 = x_1 = x_{n-m}$$

y je prefix x-a

$x = yw$

$ywy = yyw \qquad |w| = n - m$

$wy = yw$

Indukcija na $n+m$

$n + m = 1 \Rightarrow n = 1 \quad m = 0$

$y = \lambda = z^0 \qquad x = z$

Recimo da velja to za manjše kosed od $n+m$

$wy = yw$

1) $y = \lambda \Rightarrow |w| = |x| \Rightarrow w = x$

$y = x^0$ ✓

2) $y \neq \lambda \Rightarrow |w| < |x| \Rightarrow |y+w| < n+m \Rightarrow$

$w = z^{n-m} \; y = z^m \rightsquigarrow x = z^n \; y = z^m$

1.2) $x, y \in A^*$

Pravimo da sta konjugirani: $(x \sim y)$ če

$\exists u, v \in A^*.\ x = uv,\ y = vu.$

$x \sim y \Leftrightarrow \exists z.\ xz = zy$

$(\Rightarrow)$ $x \sim y$ $(x = uv\ \ y = vu)$

$uvu = uy$

$\parallel$

xu torej $\exists u \in A^*.\ xu = uy.$

$(\Leftarrow)$ Recimo da $xz = zy$

Potem $|x| = |y|$

1) $|z| \leq |x|$

$\Rightarrow z$ je prefix x

$x = z \cdot w \rightsquigarrow$

$\Rightarrow z$ je postfix y

$y = rz$ $\qquad zwz = zrz$

$\Rightarrow w = r$

$\Rightarrow x = zw \quad y = wz$

2) $|z| > |x|$

$\Rightarrow x$ je prefix z $\quad z = xw$

$\Rightarrow y$ je postfix z $\quad z = ry$

$xry = xwy \quad \rightsquigarrow r = w$

$xz = xxw = ryy$

$\vdots$

$xz = xxx \ldots xw_n = r_n y \ldots y = zy$

Potenje w_n

$\vdots$

$x \sim y \quad y \sim z \quad z \sim w$

$\exists a \quad xa = ay$

$\exists b \quad yb = bz$

$\exists c \quad zc = cw$

$xab = ayb = bz$

$xabc = bzc = bcw$

2.1) $a \in A \quad L \subseteq A^*$

$$a^{-1}L^* = a^{-1}\bigcup_{n=0}^{\infty} L^n = a^{-1}(\{\lambda\} \cup L \bigcup_{n=2}^{\infty} L^n) =$$

$$= \emptyset \cup a^{-1}L \cup a^{-1}\bigcup_{n=2}^{\infty} L^n =$$

$$= a^{-1}L \cdot a\{u \quad . \quad au \in \bigcup_{n=2}^{\infty} L^n\} =$$

$$= a^{-1}L\{au \in \bigcup_{n=1}^{\infty} ; \} =$$

$$= a^{-1}L \cdot (\emptyset \cup a^{-1}\bigcup_{n=1}^{\infty} L^n) = (a^{-1}L)\cdot L^*$$

2.2)

a) $X^{-1}(L_1 + L_2) = X^{-1}L_1 + X^{-1}L_2$

$$\bigcup_{a \in X} a^{-1}(L_1 + L_2) = \bigcup_{a \in X}(a^{-1}L \cup a^{-1}L_2) =$$

$$= \bigcup_{a \in X} a^{-1}L \quad \cup \bigcup_{a \in X} a^{-1}L_2 = X^{-1}L_1 + X^{-1}L_2$$

b) $(XY)^{-1}L = Y^{-1}(X^{-1}(L))$

$$\bigcup_{\substack{a\in X\\ b\in Y}} (ab)^{-1}L = \bigcup_{\substack{a\in X\\ b\in Y}} b^{-1}(a^{-1}(L)) = \bigcup_{a\in A} y^{-1}(a^{-1}L) = y^{-1}(X^{-1}L)$$

c) $(X^*)^{-1}L = (x^*)^{-1}(x^{-1}L)+L$

$$\left(\bigcup_{n=0}^{\infty} X^*\right)^{-1} L = \bigcup_{a\in X^*} a^{-1}L = \bigcup_{a\in\sum_{n=1}^{\infty} X^n} a^{-1}\bigcup_{b\in X} b^{-1}L =$$

$$= \bigcup_{a\in\sum_{n=1}^{\infty} X^n} a^{-1}(X^{-1})L = U(Xa)^{-1}L$$

$$(X^*)^{-1}L = \sum_{a\in X^*} a^{-1}X = L + \sum_{a\in X^+} a^{-1}L =$$

$$= L + \sum_{a\in X}\sum_{v\in X^*}(av)^{-1}L = L + \sum_{v\in X^*} v^{-1}(X^{-1}L) =$$

$$= (X^*)^{-1}(X^{-1}L) + L$$

$\varphi: A^* \longrightarrow B^*$ $\quad L \subseteq A^*$ racionalen

$\varphi(L) \subseteq B$ je racionalen

$\varphi(\emptyset) = \emptyset$

$\varphi(a) \in B^*$

1) $\varphi(L_1 + L_2) = \varphi(L_1) + \varphi(L_2)$
2) $\varphi(L_1 L_2) = \varphi(L_1)\varphi(L_2)$
3) $\varphi(L^*) = \varphi(L)^*$

1) $\varphi_*(\{u;\ u \in L_1 \cup L_2\}) = \{\varphi(u);\ u \in L_1 \cup L_2\} =$
$= \varphi_*(L_1) \cup \varphi_*(L_2)$

2) Podobno

3) $\varphi(\{u;\ u \in \bigcup_{n=0}^{\infty} L^n\}) = \varphi_*(L)^*$

5)

a) 0^*ab1^*

$0 = \emptyset \qquad 0^* = 1 + 0 + 0^2 + \ldots = \{\lambda\}$

$1 = \{\lambda\}$

$1ab1 = ab$

b) $((ab\,b)^*)^* = \left(\sum_{n=0}^{\infty}(abb)^n\right)^* =$

$= \sum_{k=0}^{\infty}\left(\sum_{n=0}^{\infty}(abb)^n\right)^k =$

$\neq \sum_{k=0}^{\infty}\sum_{n=0}^{\infty}\binom{n}{k}(abb)^{n-k} = \sum_{n=0}^{\infty}(abb)^n = (abb)^*$

c) $a^*b + a^*ba^* = a^*b(1+a^*) = a^*ba^*$

$a^*b + a^*ba^* = a^*b(1+a^*) = a^*ba^*$

$(a^*b)^*a^* =$

$\left(\sum_{n=0}^{\infty}a^*b\right)a^* = \sum_{n=0}^{\infty}a^*ba^* = \sum_{n=0}^{\infty}\sum_{k=0}^{\infty}a^*b\sum_{k=0}^{\infty}a^k =$

$= \sum_{n=0}^{\infty}(\lambda\cdot b + a^+b)\cdot(b\cdot\lambda + b\cdot a^+)$

$(a+b)^* = \sum_{n=0}^{\infty}a+b = \sum_{n=0}^{\infty}a + \sum_{n=0}^{\infty}b = a^*+b^*$